Assignment - I

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1) Types of Normalization techniques in deep learning.

→ Normalization is a crucial techique in deep learning used to stabilize training, reduce internal covariate shift, improve gradient flow, and speedup convergence. Different normalization methods serve different architectures and data shapes.

1. Batch Normalization

Batch normalization normalizes the input of each layer across the batch dimension. It computes the mean and variance of each feature across all samples in a mini-batch.

Formula:

$$\mu_B = \frac{1}{m} \sum_{i=1}^{m} x_i$$

$$\hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}$$

$$y_i = \gamma \hat{x}_i + \beta$$

Advantages

- Faster training
- Reduces internal covariate shift
- Allows higher learning rates
- Adds slight regularization effect.

Disadvantages

- Dependent on batch size
- Fails when batch size is small.
- Harder to use in RNNs

Used in

- CNN architectures (ResNet, VGG variants)
- old Transformer versions.

2. Layer Normalization

LayerNorm normalizes across the features (channels of a single sample instead of across the batch.

Formula

$$\mu = \frac{1}{H}\sum_{i=1}^{H} x_i, \quad \sigma^2 = \frac{1}{H}\sum_{i=1}^{H}(x_i-\mu)^2$$

Advantages

- works well with small batch sizes.
- Ideal for sequence models.
- Better stability for RNNs and Transformers

Used in

- BERT
- GPT models
- Transformer encoder-decoders
- RNNs / LSTMs

3. Instance Normalization

InstanceNorm normalizes each channel of each sample individually.

It is like BatchNorm but computed separately for each instance and channel.

Advantages

- Removes instance-specific contrast.
- Great for style transfer (makes style independent of image contrast).

Used in

- style transfer networks
- Some GAN architectures

4. Group Normalization

GroupNorm divides channels into groups and Normalizes within each group.

Example: 32 channels grouped into 4 groups of 8 channels each.

Advantages

- Independent of batch size
- More stable than BatchNorm for small batch sizes.

- Good balance between LayerNorm and Instance Norm.

Used in

- object detection models
- segmentation models (Mask R-CNN)

5. Weight Normalization

Instead of Normalizing activations, WeightNorm normalizes the weights of neurons.

It decomposes weights as:

$$w = \frac{g}{\|v\|} v$$

Advantages

- Helps optimizer coverge faster
- Makes training more stable in RNNs.

Disadvantages

- Less common than BatchNorm / LayerNorm.
- Doesn't help as much in CNNs

Used in

- Custom RNNs
- Reinforcement learning networks.

6. Spectral Normalization

SpectralNorm Normalizes the largest singular value of the weight matrix.

$\beta = \frac{\lambda}{\|\nabla D\|}$

λ controls the Lipschitz constant of the network
preventing gradients from exploding.

Advantages
- Extremely useful for GANs
- stabilizes discriminator training.

Used in
- WGAN-GP
- SAGAN
- BigGAN

7. Pixel Normalization

PixelNorm Normalizes each pixel across all channels.

Formula

$$x_{norm} = \frac{x}{\sqrt{\frac{1}{N}\sum x^2 + \epsilon}}$$

Advantages
- Used in GANs to replace batchnorm.
- Eliminates dependency on both batch size and channel statistics.

Used in
- Progressive GANs (PGGAN)

fference between keras, tensor flow, Pytorch.

Deep learning frameworks help build, train, and deploy machine learning models.

The three most widely used frame works are:

1. keras
2. Tensor flow
3. Pytorch

Keras

Keras is a high-level deep learning API designed for fast prototyping and easy model building.

originally it was a standalone library, but now it is integrated into TensorFlow as:

characteristics

- simple and user-friendly
- Minimal code required
- Good for beginners
- Provides abstraction over complex operations

strengths

- very easy to write models
- clean and readable syntax.
- Rapid prototyping
- Good documentation

sses
control over lower-level operations
ot ideal for highly customized architectures.
. slower debugging because of abstraction.

Best for
. Beginners in deep learning
. Quick model development
. standard CNN, RNN, DNN architectures.
. Academic projects / small prototypes.

2. Tensor flow
Tensor flow is a full-scale deep learning framework and ecosystem developed by Google.
keras is built on top of it.

Tensor flow provides:
. Low-level operations
. High-level APIs
. Production deployment tools
. Mobile / edge support (Tensorflow lite)
. Browser support (Tensorflow.js)
characteristics
. scalable and flexible
. suitable for both research and production
. supports distributed training.

strengths

- Runs on GPU/TPU
- supports deployment in mobile, web and servers
- Strong community support.
- Great for large industry projects.
- TensorBoard for visualization

weaknesses

- More complex than keras
- Lower flexibility compared to Pytorch for research.
- steeper learning curve.

Best for

- Industry/enterprise applications
- Deploying ML models to productions
- large-scale training
- Mobile AI applications

3. PyTorch

- PyTorch is a deep learning framework developed by facebook (Meta).
- Unlike TensorFlow (which originally used static graphs), PyTorch uses dynamic computation graphs, making it more intuitive.

characteristics

- Pythonic and clean

- very Popular in research
- Easy debugging
- Dynamic graph execution

strengths

- Most flexible for custom architectures.
- Easy to debug using native Python tools.
- Preferred for NLP and computer vision research
- Acts like regular Python code.
- Dynamic computation graph.

weaknesses

- Deployment support used to be weaker.
- fewer tools for mobile/web compared to Tensorflow
- steeper learning curve for beginners.

Best for

- Research and experimentation
- Custom and complex model architectures
- NLP, Transformers, GANs
- Academic papers and prototypes.